ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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March 2026 | Confidential — Investor Data Room

1. EXECUTIVE SUMMARY

The global hydrogen market is **$225 billion** and growing, but the electrolysis segment is at an inflection point defined by a stark reality: **less than 1% of the world's 100 Mt annual hydrogen production is low-emission**, and the industry's cost trajectory has moved *against* green hydrogen over the past two years. BNEF tripled its long-term green hydrogen cost forecast in December 2024. Capex overruns are systemic (NEOM: +68%). A wave of project cancellations — Ørsted, Air Products ($4.5B), bp Australia (26 GW) — has cleared the field of undifferentiated players.

This creates a structural opening for a technology that fundamentally changes the cost equation.

Key takeaways for investors:

- **Tobe's LCOH of $1.81/kg at $0.04/kWh sits 64–72% below the institutional consensus** of $4.50–$6.50/kg for current renewable electrolysis — and does not depend on tax credits, waste heat, or sub-market electricity pricing.
- **The 45V Production Tax Credit ($3/kg for 10 years) is in force** with a construction-start deadline of December 31, 2027. No legal challenges filed. With 45V, Tobe's effective hydrogen cost approaches near-zero.
- **The addressable on-site electrolysis market is $12.7B today**, growing to **$25B by 2035**, representing 65–75% of all hydrogen consumption produced at point of use.
- **Incumbent competitors are in distress.** Plug Power faces existential liquidity risk ($8.2B accumulated deficit). Nel revenue fell 31%. No OEM has commercially demonstrated sub-$2/kg LCOH without subsidies.
- **$3M–$15M+ in non-dilutive funding** is accessible within 12–24 months via federal grants, state incentives, and tax credits — preserving equity value during the seed stage.
- **First deployment at Zeeco ARC (Q2 2026)** positions Tobe as the first membrane-free electrolyzer to reach commercial operation among the next-generation cohort.

LCOH Comparison Chart

2. MARKET OVERVIEW

Total Addressable Market

METRIC	VALUE	SOURCE
Global hydrogen market (2025)	**$225B**	MarketsandMarkets
Projected market (2030)	**$312B**	MarketsandMarkets
Projected market (2040)	**$761B**	ResearchAndMarkets
Long-term addressable (2050)	**~€10 trillion**	Goldman Sachs
Global annual hydrogen demand	**~100 Mt**	IEA Global Hydrogen Review 2024
Low-emission share of production	**<1%** (~0.8 Mt)	IEA

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Serviceable Addressable Market: On-Site Electrolysis

Tobe targets the distributed/captive hydrogen production segment — the dominant mode of hydrogen consumption globally.

SEGMENT	MARKET SIZE	GROWTH	RELEVANCE
Captive hydrogen generation (all methods)	**$138.4B** (2024)	Stable	Grey-to-green conversion opportunity
Captive electrolysis specifically	**$12.7B** (2025)	**→ $25B by 2035**	Tobe's primary market
On-site share of total H_2 production	**65–75%**	Increasing	Eliminates $2–7/kg transport cost

Distributed electrolysis eliminates ~70% of hydrogen transport costs, which is the single largest barrier to green hydrogen adoption in industrial settings. Tobe's modular, containerized architecture (T-50 and T-125 skids) is purpose-built for this market.

Electrolyzer Market Dynamics

METRIC	2024 VALUE
Installed global electrolyzer capacity	**~2 GW operational**
Annual deliveries (record year)	**3.2 GW** (2x year-over-year)
Global manufacturing capacity	**38–41 GW/yr** (China >20 GW)
Electrolyzer market value	**$3.75B**
Projected market (2030)	**$14–78B** (range reflects uncertainty)

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Critical imbalance: China's manufacturing capacity of >20 GW/yr significantly exceeds global demand of ~2 GW, creating pricing pressure on conventional alkaline and PEM technologies. Tobe's differentiated architecture avoids this commodity trap.

Growth Trajectory & Demand Drivers

Market Size Trajectory

Primary demand drivers:

1. **Decarbonization mandates** — Refining (43 Mt, 44% of demand) and ammonia/fertilizer (35 Mt, 35%) face increasing pressure to replace grey hydrogen
2. **The 45V PTC** — Up to $3/kg for 10 years makes green hydrogen cost-competitive with grey in optimal conditions
3. **Energy security** — On-site production eliminates dependence on pipeline infrastructure and volatile natural gas pricing
4. **Industrial process heat** — Data center backup, steel DRI, and methanol represent emerging demand vectors

Green hydrogen growth: 60% year-over-year in 2024, projected to reach **88.6% of low-carbon hydrogen by 2030** (GlobalData). The transition from grey to green is not a question of if, but when — and at what cost.

Regional Investment Commitments (FID+, 2025)

REGION	COMMITTED INVESTMENT	COMMITTED CAPACITY
China	**$33B**	1.6 Mtpa
North America	**$23B**	2.6 Mtpa
Europe	**$19B**	0.7 Mtpa
India	**$14B**	0.3 Mtpa
Middle East	**$11B**	0.5 Mtpa
Global Total	**$110B**	**6+ Mtpa**

Source: Hydrogen Council Global Compass 2025

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3. LEVELIZED COST OF HYDROGEN (LCOH) ANALYSIS

The Industry's Central Problem

Institutional consensus places current unsubsidized green hydrogen LCOH at **$4.50–$6.50/kg** (McKinsey/Hydrogen Council, Dec 2023). Real-world project data is worse: BNEF's December 2024 assessment found actual costs of **$3.74–$11.70/kg globally**, with installed capex averaging **~$3,000/kW** — 60–300% above prior modeling assumptions.

The gap between grey hydrogen (**$0.90–$1.35/kg** at current Henry Hub pricing) and green hydrogen remains the binding constraint on adoption.

Tobe's Cost Position

BENCHMARK	LCOH	TOBE AT $1.81/KG
McKinsey/HC consensus (unsubsidized)	$4.50–$6.50/kg	**64–72% below**
DOE PEM baseline (grid, current tech)	$6.70/kg	**73% below**
Lazard ALK 100MW (unsubsidized)	$4.33/kg	**58% below**
Lazard PEM 100MW (with $3/kg PTC)	$2.48/kg	**27% below the subsidized case**
BNEF global floor	$3.74/kg	**52% below**
DOE 2026 target (Wind Class 1, advanced)	$1.90/kg	**5% below at 2x the electricity price**
Grey hydrogen (2025 US)	$0.90–$1.35/kg	Within competitive range

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LCOH Comparison Chart

How $1.81/kg Is Achieved — Without Subsidies

Tobe's cost advantage derives from three structural factors:

1. Efficiency (>92% HHV vs. 55–70% conventional) Electricity accounts for **60–75% of LCOH** in every institutional model (McKinsey, DOE, IEA). Every 10 kWh/kg improvement in efficiency saves ~$0.30–$0.50/kg at typical electricity prices. Tobe's >92% HHV system efficiency means dramatically less electricity per kilogram of hydrogen than any conventional PEM or alkaline system.

2. Capex ($47/cm² vs. $5,000–$10,000/cm² conventional) Membrane-free architecture eliminates the most expensive components: no PEM membranes (Nafion), no iridium catalysts, no active cooling system. The result is a fundamentally different cost structure that does not track conventional learning curves.

3. Realistic electricity pricing ($0.04/kWh) Oklahoma industrial electricity averages **$0.036–$0.044/kWh** (EIA data). Tobe's LCOH assumes $0.04/kWh — a publicly available, contract-verifiable rate, not a hypothetical PPA. By contrast, every other sub-$2/kg OEM claim assumes electricity at $0.012–$0.035/kWh, available only in top renewable resource locations.

Third-Party Methodology Validation

Tobe's LCOH calculation uses the **S&P Global Platts-published methodology** — the same framework used by commodity traders and financial institutions globally. S&P Platts discontinued cost-based green hydrogen assessments in December 2025, pivoting to market-based pricing. **Tobe is the only electrolyzer startup with Platts-verified pricing methodology** for its LCOH claim.

Sub-$2/kg Claims: Competitive Credibility Assessment

COMPANY	CLAIM	CAVEATS	COMMERCIAL STATUS
Tobe Energy	**$1.81/kg at $0.04/kWh**	**No subsidies required**	First deployment Q2 2026
Advanced Ionics	<$1.00/kg	Requires free waste heat + 90% CF + $0.02/kWh	TRL 4–5, no commercial deployment
Hysata	<$1.50/kg	Unspecified RE PPA; lab-verified only	No commercial deployment
Nel Hydrogen	$1.50/kg (2021 target)	**Target missed entirely**	Revenue down 31%
Ohmium	$1.00/kg (India)	Contingent on Indian government incentives	Not achieved
Plug Power	$2.00/kg (CEO verbal claim)	No supporting data; actual plant losses suggest $4–12/kg	Facing potential bankruptcy

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No large-scale project globally has publicly disclosed LCOH below $3/kg. The only available figures are analyst estimates. Tobe's claim is unique in being (a) pre-subsidy, (b) at publicly available electricity prices, and (c) validated against institutional methodology.

Regional Hydrogen Pricing Context

Regional Pricing

MARKET	GREY H_2 (SMR)	GREEN H_2 (ELECTROLYSIS)	TOBE ADVANTAGE
Texas Gulf Coast (pipeline)	$0.70–$1.32/kg	$3.00–$5.00/kg	Immediately competitive at $1.81/kg
Oklahoma (greenfield)	$1.20–$2.00/kg (est.)	$4.00–$7.00/kg	**Undercuts grey hydrogen**
California (retail)	N/A	$24.99–$32.94/kg	Massive margin opportunity
Northeast US	$1.80–$4.00/kg	$6.00–$12.00/kg	Significant cost advantage

Cost Trajectory Projections

SOURCE	2030 GREEN H_2 PROJECTION	TOBE POSITION
McKinsey/Hydrogen Council (Dec 2023)	$2.50–$4.00/kg	Already below 2030 floor
BNEF (Dec 2024, revised)	$1.60–$5.09/kg	At low end of 2050 projection — today
IEA NZE Scenario	$2.00–$9.00/kg	Below best case
DOE Hydrogen Shot	$1.00/kg by 2031	Within striking distance

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Investor consensus range: $2–5/kg globally by 2030; $1.50–$3.50/kg by 2035 for dedicated renewable electrolysis. Tobe is already at the low end of the 2035 projection in 2026.

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4. COMPETITIVE LANDSCAPE

Technology Comparison

Competitive Landscape

PARAMETER	CONVENTIONAL PEM	CONVENTIONAL ALKALINE	SOEC	TOBE (MEMBRANE-FREE)
System efficiency (HHV)	55–65%	60–70%	70–85%	**>92%**
Operating temperature	50–80°C	60–80°C	600–850°C	**<30°C (near-ambient)**
Capex ($/cm²)	$5,000–$10,000	~$2,000	$4,000–$5,000	**$47**
Installed capex ($/kW)	$2,000–$3,000	$1,700–$2,200	$4,000–$5,000	**Fundamentally lower**
Membranes required	Yes (Nafion)	Yes (diaphragm)	Yes (ceramic)	**None**
Rare earth metals	Yes (Ir, Pt)	No	Yes (various)	**None**
Cooling system	Required	Required	N/A (high-temp)	**None**
Footprint	Baseline	~1.5x PEM	~0.5x PEM	**5x smaller**
BABA compliance	Challenging	Moderate	Challenging	**95% US-sourced**

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Next-Generation Competitor Profiles

COMPANY	TECHNOLOGY	TOTAL RAISED	TRL	KEY METRIC	RISK FACTOR
Electric Hydrogen	Advanced PEM	**$600M+**	8–9	$3.90/kg pre-PTC; 1.2 GW/yr factory	Conventional PEM architecture at scale
Ohmium	Modular PEM	**$350M+**	8–9	2 GW/yr gigafactory; 400 MW NTPC deal	Still PEM-based; iridium dependence
Hysata	Capillary-fed ALK	**>$150M**	6–7	41.5 kWh/kg (95% system efficiency)	No commercial deployment; Australian
Verdagy	Advanced AWE	**~$180M**	7–8	GW-scale factory shipping	Incremental improvement on alkaline
Advanced Ionics	Steam electrolyzer	**~$20M**	4–5	<35 kWh/kg electrical (requires waste heat)	Closest technology analog; conditional claims
Sunfire	SOEC	**>€500M**	9	37.5 kWh/kg; largest SOEC deployed	600–850°C operation; ceramic fragility
Topsoe	SOEC	Public co.	8–9	500 MW/yr factory; 20–30% better than PEM	High-temperature complexity

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Incumbent Distress Creates Opportunity

- **Plug Power:** $1.7B net loss in FY2025; **$8.2B accumulated deficit**; DOE $1.7B loan suspended; 10-K explicitly contemplates bankruptcy
- **Nel Hydrogen:** Revenue **down 31%** in FY2025; NOK 799M in impairments; next-gen platform not launching until May 2026
- **Air Products:** Exited $4.5B Texas green hydrogen JV; strategic retreat from green hydrogen
- **bp:** Exited 26 GW Australian renewable energy project; hydrogen ambitions scaled back

Tobe's Differentiation

Why membrane-free matters for investors:

1. **No iridium supply chain risk.** PEM electrolyzers depend on iridium (~$5,000/oz, produced primarily in South Africa). Tobe uses zero rare earth metals.

2. **No membrane degradation.** PEM membranes (Nafion) degrade over 40,000–60,000 hours, requiring costly stack replacements at 11–35% of installed capex. Tobe eliminates this recurring cost.
3. **No cooling system.** Near-ambient operation (<30°C) eliminates the cooling infrastructure that adds 15–25% to conventional system costs and complexity.
4. **BABA-compliant by design.** 95% US-sourced materials, manufactured in Oklahoma. No waiver needed. Positions Tobe for every federal procurement requirement.
5. **5x smaller footprint.** Enables deployment in space-constrained industrial settings where conventional electrolyzers cannot fit.

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5. REGULATORY & INCENTIVE ENVIRONMENT

Section 45V Clean Hydrogen Production Tax Credit

The **45V final regulations were published January 10, 2025** and are in force. This is the single most important policy enabler for US green hydrogen economics.

PARAMETER	DETAIL
Maximum credit	**$3.00/kg for 10 years**
Qualification threshold	<0.45 kg CO_2e/kg H_2 (Tier 1)
Construction start deadline	**December 31, 2027** (OBBBA)
Legal challenges	**None filed**
Three pillars	Incrementality (36-month COD window), temporal matching (annual through 2029, hourly from 2030), deliverability (same DOE grid region)

Impact on Tobe: At $1.81/kg pre-subsidy, the full $3/kg 45V credit would bring Tobe's effective LCOH to approximately **negative $1.19/kg** — creating a margin structure unprecedented in hydrogen production. Even partial credit qualification yields immediate cost competitiveness with grey hydrogen.

Inflation Reduction Act — Broader Provisions

PROVISION	STATUS	RELEVANCE TO TOBE
§48C Manufacturing ITC (30%)	Fully allocated at $10B; closed	Prepare for any future round
§45X Advanced Manufacturing	Active; electrolyzers not yet listed	If expanded, $2M–$10M+/yr value
§41 R&D Tax Credit (payroll offset)	**Immediately actionable**	**$210K–$420K/yr** against payroll taxes
Bonus Depreciation (§168)	100% restored by OBBBA	**$2M–$10M+ first-year deduction**
§179D Energy-Efficient Buildings	Active; construction before Jun 30, 2026	~$297K deduction for 50K sq ft facility
Opportunity Zones	Active; OZ 2.0 permanent under OBBBA	Lowers investor cost of capital

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DOE Hydrogen Shot

The DOE's $2/kg by 2026 interim target was **not met** on an unsubsidized basis by the broader industry. Current PEM LCOH remains $5–7/kg before credits. The $1/kg by 2031 ultimate target requires simultaneously achieving electricity ≤$0.012/kWh, capex ≤$150/kW, and capacity factor ≥74%.

Tobe is currently the only US company demonstrating a pathway to the DOE's interim target using commercially available electricity.

DOE H2Hubs: Dismantled but Relevant

All 7 DOE-funded H2Hubs have been targeted for termination under the current administration (~$7.56B canceled). California leads a 13-state lawsuit. However, private participants within hub frameworks continue procurement — registering as an OEM supplier with ARCHES, HyVelocity, and Heartland remains a viable channel regardless of federal funding status.

State-Level Incentives (Oklahoma Focus)

INCENTIVE	VALUE	STATUS
Manufacturing Sales Tax Exemption	**$200K–$800K** on equipment	Active — apply immediately
Five-Year Ad Valorem Exemption	**$100K–$450K/yr** property tax savings	Active
Quality Jobs Program	**5% of payroll, 7 years** (~$70K/yr)	Active
Alternative Fueling Infrastructure Credit	**45% of H_2 infrastructure cost**	Active through 2028
OIEP (Innovation Expansion Program)	**$150K payroll rebate**	Opens March 30, 2026
R&D Rebate Program (SB 324)	**5% rebate on OK-based QREs** ($75K–$150K/yr)	Active

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International Context

- **EU:** €18.9B in public funding via four IPCEI waves; Hydrogen Bank auctions awarding €0.33–€1.88/kg support
- **Australia:** A$2/kg Hydrogen Production Tax Incentive enacted February 2025
- **Japan:** $19.2B CfD scheme for hydrogen/ammonia imports
- **South Korea:** World's first clean hydrogen power market mandate (2,600 GWh for 2026)

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6. NON-DILUTIVE FUNDING PIPELINE

Analysis of 60+ programs identifies **$3M–$15M+ in near-term non-dilutive capital** accessible within 12–24 months, with **$500K–$1.75M+ in annual recurring value** from tax incentives alone.

Top-Priority Opportunities

PRIORITY	PROGRAM	VALUE	DEADLINE	FIT
1	**ARPA-E SPARKS** (DE-FOA-0003164)	Up to $500K	Rolling — open now	Membrane-free electrolysis is a direct fit for transformational energy R&D
2	**Section 41 R&D Credit** (Payroll Offset)	$210K–$420K/yr	Next tax filing	Immediately actionable as QSB
3	**OIEP (Oklahoma)**	$150K payroll rebate	Mar 30–Apr 10, 2026	Urgent window
4	**TERA-Award 2026**	Up to $1M prize	April 22, 2026	Low effort, high upside
5	**Army SBIR — On-Demand H_2**	Up to $2M (Direct-to-Phase-II)	Rolling	Military hydrogen generation is an explicit topic
6	**Breakthrough Energy Fellows**	$50K–$250K	Rolling	Strong fit for Explorer track
7	**OK Manufacturing Sales Tax Exemption**	$200K–$800K	Apply now	Immediate savings on equipment
8	**Bonus Depreciation (§168)**	$2M–$10M+ deduction	On tax return	100% first-year equipment write-off

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Estimated Non-Dilutive Value Summary

CATEGORY	NEAR-TERM (YEAR 1–2)	ANNUAL RECURRING
Federal Grants	$500K–$2.5M	—
Federal Tax Credits & Deductions	$2M–$10M+	$210K–$500K/yr
Oklahoma State Programs	$250K–$1M	$300K–$1.25M/yr
Private Foundations & Prizes	$50K–$1.5M	—
Total	**$3M–$15M+**	**$500K–$1.75M+/yr**

Political Context

Federal hydrogen funding is significantly constrained under the current administration. DOE HFTO FOAs are frozen, H2Hubs have been canceled, and SBIR/STTR experienced a 5-month lapse (reauthorized February 2026). **ARPA-E SPARKS and DoD SBIR remain the most viable open federal pathways.** State programs and tax incentives are unaffected and represent the strongest near-term value.

This dynamic favors Tobe: competitors dependent on federal grants face delays and uncertainty. Tobe's economics work without subsidies, and the non-dilutive funding pipeline serves as capital efficiency amplification rather than a dependency.

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7. KEY RISKS & MITIGANTS

RISK	SEVERITY	MITIGANT
Technology scale-up risk — First deployment at Zeeco ARC is pre-commercial	High	Zeeco partnership provides EPC capabilities and industrial validation environment. Modular skid architecture (T-50, T-125) limits integration risk.
45V PTC uncertainty — Construction start deadline of Dec 31, 2027; future administration could modify	Medium	Tobe's $1.81/kg does not depend on 45V. The credit is upside, not survival. Unlike EH2 ($3.90/kg pre-PTC), Tobe is viable without subsidies.
Offtake gap — Only 3.6 Mtpa of global binding offtake exists vs. 6 Mtpa committed supply	High	Distributed/on-site model eliminates offtake intermediary. Target captive industrial users with existing grey hydrogen consumption. $100M pipeline already identified.
Chinese electrolyzer pricing pressure — Installed alkaline at $600–$1,300/kW	Medium	Tobe competes on efficiency and BABA compliance, not commodity pricing. Chinese systems cannot access 45V or federal procurement channels.
Grey hydrogen remains cheap — $0.90–$1.35/kg at current Henry Hub	Medium	Carbon pricing trajectory, regulatory mandates, and ESG procurement requirements are compressing the green premium. Tobe at $1.81/kg is already within range.
Capex overrun risk — Systemic across hydrogen projects (NEOM +68%)	Low–Medium	Tobe's membrane-free architecture eliminates major cost drivers (membranes, cooling, rare earths). 100% in-house manufacturing in Oklahoma. 5x smaller footprint reduces civil works.
Demand timing uncertainty — Steel DRI stalled, hub cancellations, slow adoption	Medium	Focus on near-term markets with proven demand: ammonia, refining, industrial process, forklifts. Avoid speculative demand sectors.
Efficiency claims require independent validation	Medium	Platts-verified methodology for LCOH. INL-style independent testing is a planned milestone post-deployment. Zeeco ARC provides metered, auditable operating data.
Team and execution risk — 8+ FTE, pre-revenue	Standard for stage	$250M+ combined energy infrastructure deployment experience. Manufacturing facility operational. Techstars NYC and Cortado-backed.
Fundraising environment — Cleantech capital constrained; hydrogen sector sentiment negative	Medium	Negative sentiment is a contrarian signal. Competitor distress (Plug, Nel) creates buyer's market for talent and

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RISK	SEVERITY	MITIGANT
		partnerships. $10M seed is modest relative to value creation.

The Contrarian Case

The hydrogen sector's current distress — project cancellations, cost overruns, incumbent failures — is a feature for Tobe, not a bug. The companies failing are using conventional architectures at conventional costs. The market is learning that **incremental improvements to PEM and alkaline cannot close the green premium.** Tobe's membrane-free, rare-earth-free, cooling-free architecture represents a step-function change, not an incremental one.

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The DOE's own models show sub-$2/kg is physically achievable with advanced technology at reasonable electricity prices. Tobe demonstrates this today — at commercially available Oklahoma electricity rates, using a Platts-verified methodology, without depending on tax credits that may or may not survive political cycles.

The question is not whether the hydrogen market will materialize. At 100 Mt of annual demand and $225B in value, the market exists today. The question is whether next-generation electrolysis can capture share from grey hydrogen. At $1.81/kg, Tobe's answer is yes.

This report synthesizes data from 36+ institutional sources, 10 OEM publications, 13 real-world project datasets, and 4 peer-reviewed academic papers (2024–2026). All dollar figures cite the original source's dollar-year basis. Primary source documents are available upon request for due diligence verification.

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